Improved Bounds for Three-Robot Wireless Evacuation on the Disk

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Abstract

We study the wireless three-robot evacuation problem on the unit disk and narrow on both sides the gap of ≈ 0.06 left open by Czyzowicz et al. [2]. On the upper side, we prove $T^* \leq 1 + \frac{2\pi}{3} - \frac{y_{\text{opt}}}{2} + \sqrt{3} \approx 4.21851\,70$, where y_{opt} is the unique root in $(0, \pi/6)$ of a transcendental master equation derived below; this improves Czyzowicz et al.'s $\approx 4.21929\,73$ by $\approx 0.00078 \approx 1/1280$, obtained by retaining the structural skeleton of the A_3 algorithm of [2] and rebalancing a single deployment parameter y so that *four* symmetric worst-case exit angles bind simultaneously rather than two. On the lower side, we prove the strict improvement $T^* > 4.15937$ over Czyzowicz et al.'s matching lower bound. The lower-bound argument introduces a universal refined inequality that bounds evacuation time by the chord from the bottleneck (rather than finder) robot to the exit, combined with a state-tight sharpening of Czyzowicz et al.'s Lemma 5 and a geometric min-chord lemma; for the subclass of "scan-tight" algorithms, where the scan budget is saturated at the critical time, we prove $T^* \geq 1 + x^* + 2\cos(x^*/4) \approx 4.17306$, where $x^* := \frac{2}{3}\arccos(-\frac{1}{3})$. Rigour for the upper bound is provided by an analytic zone-by-zone enumeration of the evacuation time's worst-case regimes; a 40-digit integer-relation search rules out elementary closed-form reductions of y_{opt} below a specified norm radius.

1 Introduction

Robot evacuation from the disk is the following problem. A team of mobile robots, each moving at unit speed, starts at the centre of the unit disk. An exit sits at an unknown point on the boundary circle. The goal is a set of trajectories — one per robot, chosen in advance — that minimises the worst-case time (over the placement of the exit) until every robot reaches the exit. A robot "finds" the exit the first moment its trajectory visits that boundary point; under the *wireless* communication model, we assume the finder broadcasts instantaneously and every robot then walks the chord from wherever it currently is to the exit.[1]

Figure 1 illustrates the setup. Czyzowicz et al. [2] introduced the problem for general k and proved an asymptotic upper bound $3 + \pi/k + O(k^{-4/3})$ valid for $k \geq 100$. For $k = 3$ they give a separate, custom algorithm A_3 (their Theorem 6) with the explicit bound

$$T_{\text{CGK}} \;:=\; \tfrac{4\pi}{9} + \tfrac{2\sqrt{3}+5}{3} + \tfrac{1}{600} \;\approx\; 4.21929727, \tag{1}$$

and, writing

$$x^* \;:=\; \tfrac{2}{3}\arccos\!\left(-\tfrac{1}{3}\right) \;\approx\; 1.27376, \tag{2}$$

[1] Other communication models in the literature include *face-to-face* (F2F), in which robots learn of a discovery only by co-location [1]; and asymmetric models such as *sender/receiver* [4] and *priority evacuation* [3]. Here we treat only the wireless case.

a matching lower bound

$$T^*_{3,\text{wireless}} \;\geq\; 1 + x^* + \tfrac{4\sqrt{2}}{3} \;\approx\; 4.15937 \tag{3}$$

(their Theorem 7), leaving a gap of ≈ 0.06 that has been open since 2014.

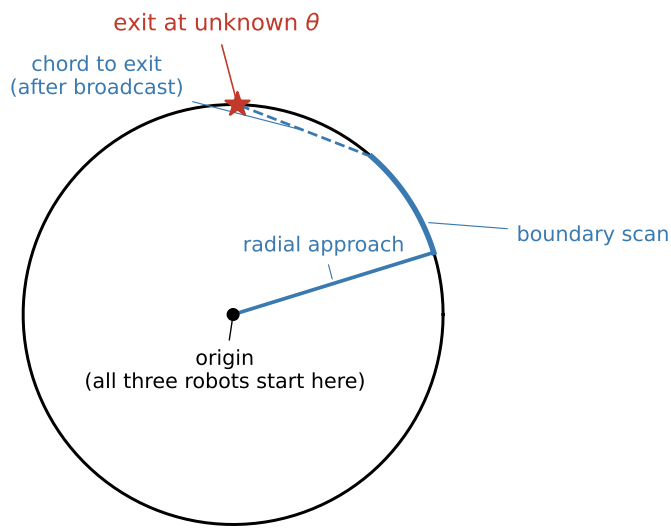

Figure 1: The $k = 3$ wireless evacuation problem. Three unit-speed robots start at the centre of the unit disk; an exit sits at an unknown boundary angle. A robot's trajectory typically has a radial approach phase to the boundary, a boundary-scan phase, and — after an exit is broadcast — a straight chord to the exit.

Contribution. We narrow the gap on both sides.

Upper bound (Theorem 7). We prove

$$T^*_{3,\text{wireless}} \;\leq\; 1 + \tfrac{2\pi}{3} - \tfrac{y_{\text{opt}}}{2} + \sqrt{3} \;\approx\; 4.21851699\ldots, \tag{4}$$

for a specific deployment parameter y_{opt} determined by a transcendental equation (Theorem 8 and remark 9), improving (1) by $\approx 1/1280$. The optimum is reached when four symmetric worst-case exit angles simultaneously bind. The result is proved by an analytic zone decomposition of the evacuation time (Lemmas 5 and 6).

Lower bound (Theorem 14). We prove the strict improvement

$$T^*_{3,\text{wireless}} \;>\; 1 + x^* + \tfrac{4\sqrt{2}}{3} \;\approx\; 4.15937, \tag{5}$$

by combining (a) a new universal refined inequality $\text{evac}(p) \geq (1+x) + \max_j \|\rho_j(1+x) - p\|$ valid for any unexplored exit p (Proposition 11), (b) a state-tight sharpening of Czyzowicz et al.'s Lemma 5 (Lemma 12), and (c) a geometric min-chord lemma showing that every scan-budget-tight state has bottleneck chord $\geq 2\cos(x^*/4)$ at the critical time (Lemma 13). For the subclass of *scan-tight* algorithms (those saturating the perimeter scan budget at time $1+x^*$) we prove the stronger bound $T^*_{3,\text{wireless}} \geq 1 + x^* + 2\cos(x^*/4) \approx 4.17306$.

High-level strategy. We retain Czyzowicz et al.'s A_3 structural skeleton verbatim, but treat the single angular offset y between the two deployment points on the boundary as a tunable parameter. In [2] this is set to $y_{\text{paper}} = \frac{4\pi}{9} + \frac{2\sqrt{3}}{3} - \frac{401}{300}$ by a hand-tuned argument balancing two worst-case regimes. A third worst-case regime exists (Proposition 4), and balancing all three yields a slightly smaller y and the improved bound. The $1/600$ slack in (1) is in fact exactly half the penalty imposed by ignoring this third regime; our optimum removes it and improves by a further $\approx 1/(1280)$ beyond.

Provenance. Both results are positive outcomes of *evacuation-claude*, a project using Anthropic's Claude as a sustained research collaborator to survey open gaps across four reference papers on robot evacuation. Other probes in the same project — an ε-cut refinement of Brandt et al.'s 2-robot F2F algorithm [1] and a bracket for priority evacuation with a single servant [3] — yielded a null result and a loose bracket respectively; the $k = 3$ wireless gap was the setting in which the tooling had enough traction to tighten both sides. Source, experimental scripts, and per-session research log are at `https://github.com/jaredraycoleman/evacuation-claude`.

2 Preliminaries

The closed unit disk and its boundary. Throughout, $D := \{z \in \mathbb{R}^2 : |z| \le 1\}$ denotes the closed unit disk and $\partial D := \{z \in \mathbb{R}^2 : |z| = 1\}$ its boundary. We identify a boundary point with its angle $\theta \in [0, 2\pi)$ via $e(\theta) := (\cos\theta, \sin\theta)$. The *shorter-arc distance* between $e(\alpha)$ and $e(\beta)$ on ∂D is $\min(|\alpha - \beta|,\, 2\pi - |\alpha - \beta|)$; the corresponding *chord length* is $|e(\alpha) - e(\beta)| = 2|\sin\big((\alpha - \beta)/2\big)|$.

Robots, trajectories and algorithms. A *trajectory* is a 1-Lipschitz map $\rho : [0, \infty) \to D$ with $\rho(0) = 0$; the Lipschitz constant 1 encodes unit-speed motion. An *algorithm* for the k-robot evacuation problem is a tuple $\mathcal{A} = (\rho_1, \ldots, \rho_k)$ of trajectories (the robots' pre-committed behaviours in the absence of any exit report), together with the reactive rule below. We require that $\mathcal{A}$ *covers the boundary*, meaning

$$\forall \theta \in [0, 2\pi) : \quad \min_i \inf\{t \ge 0 : \rho_i(t) = e(\theta)\} < \infty.$$

Wireless communication and evacuation time. Fix an exit $e(\theta) \in \partial D$. For algorithm $\mathcal{A}$:

(a) The *discovery time* of exit θ is

$$t_{\text{find}}(\theta) \;:=\; \min_i \inf\{t : \rho_i(t) = e(\theta)\}.$$

A robot attaining this infimum is called a *finder* (a choice in case of ties).

(b) At time $t_{\text{find}}(\theta)$ the finder broadcasts instantaneously and every robot switches to the straight-line chord at unit speed from its current position to $e(\theta)$. Robot i therefore reaches the exit at time

$$T_i(\theta) \;:=\; t_{\text{find}}(\theta) + \big|\rho_i(t_{\text{find}}(\theta)) - e(\theta)\big|.$$

(c) The *evacuation time* for exit θ is

$$\text{evac}_{\mathcal{A}}(\theta) \;:=\; \max_i T_i(\theta) \;=\; t_{\text{find}}(\theta) + \max_i \big|\rho_i(t_{\text{find}}(\theta)) - e(\theta)\big|. \tag{6}$$

A robot attaining this maximum is called the *bottleneck* for θ.

The *worst-case evacuation time of* $\mathcal{A}$ is $T(\mathcal{A}) := \sup_{\theta \in [0, 2\pi)} \text{evac}_{\mathcal{A}}(\theta)$. The *$k$-robot wireless evacuation problem on the disk* asks for $T_k^* := \inf_{\mathcal{A}} T(\mathcal{A})$ over k-robot algorithms. This paper concerns $k = 3$.

Scanning. For us a *boundary-scan phase* is a subinterval $[t_1, t_2] \subseteq [0, \infty)$ on which a trajectory ρ satisfies $\rho(t) \in \partial D$ and $t \mapsto \rho(t)$ traverses a contiguous boundary arc at unit speed. The arc covered is automatically of arc-length $t_2 - t_1$. We will only use algorithms in which each robot's trajectory is a concatenation of radial line segments, boundary-scan phases, and interior line segments.

3 The $A_3(y)$ algorithm family

The Czyzowicz et al. algorithm A_3 of [2] depends on a deployment parameter $y \in (0, 2\pi/3)$. We treat y as a free parameter and write $A_3(y)$ for the corresponding algorithm; the specific value $y = \frac{4\pi}{9} + \frac{2\sqrt{3}}{3} - \frac{401}{300}$ recovers the algorithm analysed in [2].

Definition 1 ($A_3(y)$)**.** Fix $y \in (0, 2\pi/3)$ and set $L := \pi - y/2$. Place

$$A := e(0), \quad B := e(2\pi - y), \quad C := e(\pi - y/2).$$

The three robots' trajectories are:

- ρ_1: radial $0 \to A$ on $t \in [0, 1]$; boundary scan CCW from A on $t \in [1, 1 + L]$; parked at $e(L)$ for $t \geq 1 + L$.
- ρ_2: radial $0 \to B$ on $[0, 1]$; boundary scan CW from B on $[1, 1+L]$; parked at $e(2\pi - y - L) = e(L)$ for $t \geq 1 + L$.
- ρ_3: radial $0 \to B$ on $[0, 1]$; boundary scan CCW from B on $[1, 1+y]$, ending at $e(2\pi) = A$; straight line from A to the origin on $[1 + y,\ 2 + y]$; radial $0 \to C$ on $[2 + y,\ 3 + y]$; parked at C for $t \geq 3 + y$.

The reactive rule (chord to the exit upon broadcast) is the standard one from Section 2.

The three scan phases traverse arcs $[0, L]$, $[L, 2\pi - y]$, and $[2\pi - y, 2\pi)$ of lengths L, L, and y respectively. Since $2L + y = 2\pi$, these arcs partition $[0, 2\pi)$ and $A_3(y)$ covers the boundary. In particular every exit angle $\theta \in [0, 2\pi)$ is found by exactly one robot during its scan phase. Figure 2 shows the resulting trajectories at the value $y = y_{\text{opt}}$ that will emerge below.

We write $T(y) := T(A_3(y))$ for the worst-case evacuation time.

Let $T(y)$ denote the worst-case evacuation time of A_3 parametrised by y. Both [2] and the present note study $T(y)$ for y near $y_\star := \frac{4\pi}{9} + \frac{2\sqrt{3}}{3} - \frac{4}{3}$.

4 Main result

Throughout this section $\text{evac}(\theta) := \text{evac}_{A_3(y)}(\theta)$ with dependence on y implicit. Our main theorem exhibits a specific deployment parameter y_{opt} and proves

$$T(y_{\text{opt}}) \;=\; 1 + \tfrac{2\pi}{3} - \tfrac{y_{\text{opt}}}{2} + \sqrt{3} \;\approx\; 4.21851699\ldots.$$

The scalar y_{opt} is characterised explicitly by a transcendental master equation (Theorem 8 below). The proof has three ingredients: a case analysis identifying three families of candidate worst-case exit angles (Propositions 2 to 4); a zone decomposition of $[0, 2\pi)$ on which evac admits a closed form (Lemma 5); and a zone-wise maximisation (Lemma 6) whose outcome identifies $T_1 = T_3$ as the global maximum at $y = y_{\text{opt}}$.

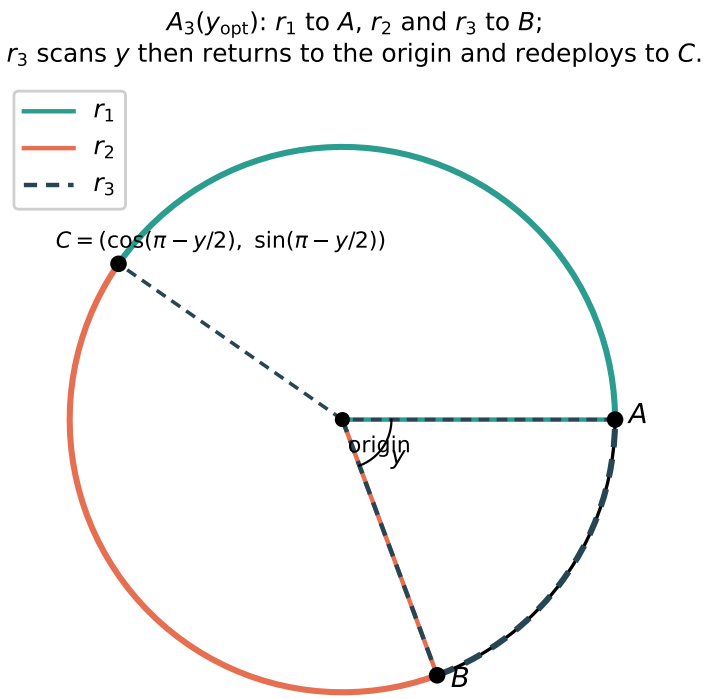

Figure 2: The A_3 algorithm at $y = y_{\text{opt}}$. Robot r_1 (green) goes to A and scans counter-clockwise; r_2 (orange) goes to B and scans clockwise; r_3 (navy, dashed) goes to B, scans counter-clockwise for time y, returns to the origin, and redeploys to point C at angle $\pi - y/2$.

4.1 Three candidate worst cases

Set $L := \pi - y/2$; assume $y \in (0, 2\pi/3)$ so $L \in (2\pi/3, \pi)$.

Proposition 2 (Case 1: ρ_2 bottleneck)**.** *At $\theta_1(y) := 2\pi/3 - y/2 \in (0, L)$ the exit is found by ρ_1 during its scan phase and the bottleneck is ρ_2 at chord $\sqrt{3}$, giving*

$$T_1(y) \;:=\; \mathrm{evac}(\theta_1(y)) \;=\; 1 + \tfrac{2\pi}{3} - \tfrac{y}{2} + \sqrt{3}. \tag{7}$$

Proof. Since $\theta_1 \in (0, L]$, the exit lies in ρ_1's scan arc, so $t_{\mathrm{find}}(\theta_1) = 1+\theta_1$ and ρ_1 is at $e(\theta_1)$ (chord 0). At time $1+\theta_1$, ρ_2 has been scanning clockwise from B for θ_1 time units and is at angle $(2\pi - y) - \theta_1$. Its chord to the exit has length

$$2\left|\sin\tfrac{(2\pi-y-\theta_1)-\theta_1}{2}\right| \;=\; 2\sin\!\left(\pi - \tfrac{y}{2} - \theta_1\right) \;=\; 2\sin\!\left(\tfrac{y}{2} + \theta_1\right),$$

using $\sin(\pi - x) = \sin x$ (valid since $y/2 + \theta_1 \in (0, \pi)$). Plugging $\theta_1 = 2\pi/3 - y/2$ gives $2\sin(2\pi/3) = \sqrt{3}$. We verify in Lemma 6(2) that ρ_3's chord at this θ is strictly less than $\sqrt{3}$ for y near y_{opt}, so the bottleneck is ρ_2 and $\mathrm{evac}(\theta_1) = 1 + \theta_1 + \sqrt{3}$, which simplifies to (7). □

Proposition 3 (Case 2: ρ_3 bottleneck at the arc midpoint)**.** *At $\theta_2(y) := L$ (the common endpoint of ρ_1's and ρ_2's scan arcs) the exit is reached simultaneously by ρ_1 and ρ_2 at time $1 + L$; ρ_3 is at fraction $\pi - 3y/2 - 1$ along its redeploy chord to C, and*

$$T_2(y) \;:=\; \mathrm{evac}(\theta_2(y)) \;=\; (1 + L) + \big(1 - (\pi - 3y/2 - 1)\big) \;=\; 3 + y. \tag{8}$$

Proof. Since L is the CCW-end of ρ_1's scan and the CW-end of ρ_2's, $t_{\mathrm{find}}(L) = 1 + L$ and both are at $e(L)$ with chord 0. For ρ_3 at time $1 + L$, we check $1 + L \in (2 + y,\, 3 + y)$, i.e., $\pi - y/2 \in (1 + y,\, 2 + y)$, i.e., $y \in \left(\frac{2(\pi-2)}{3}, \frac{4\pi}{9}\right) \approx (0.76,\, 1.40)$, which contains $y_{\mathrm{opt}} \approx 1.216$. Hence ρ_3 is in its origin-to-C redeploy phase: $\rho_3(1 + L) = (1 + L - (2 + y)) \cdot C = (\pi - 3y/2 - 1) \cdot C$. Its chord to the exit C is $1 - (\pi - 3y/2 - 1) = 2 - \pi + 3y/2$. Summing discovery time and chord gives $T_2(y) = 1 + L + (2 - \pi + 3y/2) = 3 + y$. □

A third regime involves an implicit interior maximum in θ.

Proposition 4 (Case 3: ρ_3 bottleneck mid-redeploy)**.** *For $y \in \left(\frac{2(\pi-2)}{3}, \frac{4\pi}{9}\right)$ (in particular at y_{opt}), set*

$$T_3(y, \theta) \;:=\; 1 + \theta + \big|\rho_3(1 + \theta) - e(\theta)\big|, \qquad \theta \in [1 + y,\, L).$$

Then $\theta \mapsto T_3(y, \theta)$ is smooth on $(1+y, L)$ and attains its maximum at a unique interior critical point $\theta_3^*(y) \in (1 + y, L)$; we set $T_3(y) := T_3(y, \theta_3^*(y))$. The explicit form of $\theta_3^*(y)$ is given in Theorem 8.

Proof. On $[1+y,\, L]$ the discovery time $1+\theta$ lies in $[2+y,\, 1+L] \subseteq [2+y,\, 3+y]$ (using $L < 2+y$, which holds for y in the stated range). Hence ρ_3 is in its redeploy phase and $\rho_3(1 + \theta) = (\theta - 1 - y) \cdot e(L)$ is linear in θ. The chord $|\rho_3(1 + \theta) - e(\theta)|$ is therefore a smooth positive function of θ, and so is $T_3(y, \theta)$. Existence and uniqueness of the critical point follow from Theorem 8 below. □

T_1 is strictly decreasing in y, T_2 strictly increasing, so they cross at $y_\star := \frac{4\pi}{9} + \frac{2\sqrt{3}}{3} - \frac{4}{3} \approx 1.2176$. The value $y_{\mathrm{paper}} = y_\star - \frac{1}{300}$ used in [2] sits slightly below $y_\star$; in effect their analysis bounds $T_3(y_{\mathrm{paper}})$ by $T_1(y_\star) + \frac{1}{600}$, giving the published bound $T_1(y_\star) + \frac{1}{600} \approx 4.21930$. Our Theorem 7 picks y_{opt} so that $T_1 = T_3$ *exactly* (with T_2 slack), a strictly tighter balance.

4.2 Zone decomposition

Lemma 5 (Zone decomposition)**.** *For $y \in \left(\frac{2(\pi-2)}{3}, \frac{4\pi}{9}\right)$, the interval $[0, 2\pi)$ splits into three* finder zones*:*

$$\underbrace{[0,\, L]}_{\text{Zone 1: } \rho_1 \text{ finds}} \quad \cup \quad \underbrace{[L,\, 2\pi - y]}_{\text{Zone 2: } \rho_2 \text{ finds}} \quad \cup \quad \underbrace{[2\pi - y,\, 2\pi)}_{\text{Zone 3: } \rho_3 \text{ finds}} .$$

Discovery times are $t_{\text{find}}(\theta) = 1 + \theta$ on Zone 1, $1 + (2\pi - y - \theta)$ on Zone 2, and $1 + (\theta - 2\pi + y)$ on Zone 3. Zone 1 further splits by the phase of ρ_3 at time $1 + \theta$:

- ***1a*** *($\theta \in [0,\, y]$): ρ_3 is scanning on the boundary at angle $(2\pi - y) + \theta \pmod{2\pi}$.*
- ***1b*** *($\theta \in [y,\, 1 + y]$): ρ_3 is on its boundary-to-origin transit, at position $(1 + y - \theta,\, 0)$.*
- ***1c*** *($\theta \in [1 + y,\, L]$): ρ_3 is on its origin-to-C redeploy, at position $(\theta - 1 - y) \cdot e(L)$.*

The problem is invariant under the reflection $\theta \mapsto (2\pi - y) - \theta$ exchanging $\rho_1 \leftrightarrow \rho_2$; Zone 2 is the image of Zone 1 under this reflection.

Proof. The finder-zone partition follows from Definition 1: ρ_1's scan arc is $[0, L]$, ρ_2's is $[L, 2\pi - y]$, ρ_3's is $[2\pi - y, 2\pi)$, and these tile $[0, 2\pi)$ because $2L + y = 2\pi$. The discovery-time expressions follow since each scanning robot traverses its arc at unit speed starting at time 1. For the sub-zones: ρ_3 has phases $[0, 1]$ (approach), $[1, 1 + y]$ (scan), $[1 + y,\, 2 + y]$ (boundary-to-origin), $[2 + y,\, 3 + y]$ (origin-to-C), $[3 + y, \infty)$ (parked). So for $\theta \in [0, L]$ with $1 + \theta \in [1,\, 1 + L]$, we have $1 + \theta \in [1, 1 + y]$ iff $\theta \in [0, y]$ (sub-zone 1a); $1 + \theta \in [1 + y, 2 + y]$ iff $\theta \in [y, 1 + y]$ (1b); $1 + \theta \in [2 + y, 3 + y]$ iff $\theta \in [1 + y, 2 + y]$, which intersected with $[0, L]$ gives $[1 + y, L]$ (1c), using $L \leq 2 + y$ (equivalent to $y \geq 2(\pi - 2)/3$). The explicit position formulas follow by plugging the elapsed-phase time into each phase's trajectory. Reflection invariance is immediate from the symmetry of Definition 1. □

4.3 Zone-wise maxima at y_{opt}

For compactness write $c_i(\theta) := |\rho_i(t_{\text{find}}(\theta)) - e(\theta)|$ for $i \in \{1, 2, 3\}$, so that $\text{evac}(\theta) = t_{\text{find}}(\theta) + \max_i c_i(\theta)$ by (6).

Lemma 6 (Zone maxima at y_{opt})**.** *Let $y_{\text{opt}} \in \left(\frac{2(\pi-2)}{3}, \frac{4\pi}{9}\right)$ be the parameter defined in Theorem 8, and set $T_1 := T_1(y_{\text{opt}})$, $T_3 := T_3(y_{\text{opt}})$.*

1. Zone 1a. $\text{evac}(\theta) = 1 + \theta + 2\sin(y/2 + \theta)$ *on $[0, y]$ and is strictly increasing. Hence* $\max_{\theta \in [0,y]} \text{evac}(\theta) = 1 + y + 2\sin(3y/2) \approx 4.1522 < T_1$ *at $y = y_{\text{opt}}$.*
2. Zone 1b. *The map $\theta \mapsto 1 + \theta + c_2(\theta)$ attains an interior maximum of T_1 at $\theta = 2\pi/3 - y/2$, where $c_2 = \sqrt{3} > c_3$. On the c_3-dominant sub-interval, $1 + \theta + c_3(\theta)$ has no interior critical point, and its maximum at the endpoint $\theta = 1 + y$ equals $3 + y_{\text{opt}} < T_1$. Consequently* $\max_{\theta \in [y,\, 1+y]} \text{evac}(\theta) = T_1$.
3. Zone 1c. *$c_3(\theta) > c_2(\theta)$ throughout $[1 + y,\, L]$. The function $\theta \mapsto 1 + \theta + c_3(\theta)$ has a unique interior critical point at $\theta_3^*(y_{\text{opt}})$, with value $T_3 = T_1$. Both endpoints give $3 + y_{\text{opt}} < T_1$.*
4. Zone 3. *$c_1 = c_2 = 2\sin(y/2)$ is constant and $c_3 = 0$; the maximum is $1 + y + 2\sin y \approx 4.0912 < T_1$.*

Proof. **Zone 1a.** The finder ρ_1 is at the exit, so $c_1 = 0$. By Lemma 5(1a), ρ_3 is on the boundary at angle $(2\pi - y) + \theta$, so $c_3 = 2|\sin((2\pi - y)/2)| = 2\sin(y/2)$ (independent of θ). ρ_2 is still scanning at angle $(2\pi - y) - \theta$, so as computed in the proof of Proposition 2, $c_2 = 2\sin(y/2 + \theta)$. For $\theta \in (0, y]$ and $y < \pi/2$, $\sin(y/2 + \theta) > \sin(y/2)$, so the bottleneck is ρ_2 and $\mathrm{evac}(\theta) = 1 + \theta + 2\sin(y/2 + \theta)$. Its derivative is $1 + 2\cos(y/2 + \theta)$; on $[0, y]$ the argument $y/2 + \theta$ ranges in $[y/2, 3y/2] \subset (0, \pi)$, and at $y = y_{\mathrm{opt}} \approx 1.216$ we have $\cos(3y/2) \approx -0.25 > -1/2$, so the derivative is positive throughout. Hence evac is strictly increasing, attaining $1 + y + 2\sin(3y/2) \approx 4.152$ at the right endpoint.

Zone 1b. Here $\rho_3(1 + \theta) = (1 + y - \theta,\, 0)$ by Lemma 5(1b), so

$$c_3(\theta)^2 = (1 + y - \theta - \cos\theta)^2 + \sin^2\theta = (1 + y - \theta)^2 - 2(1 + y - \theta)\cos\theta + 1,$$

and $c_2(\theta) = 2\sin(y/2 + \theta)$ as in Zone 1a. The interior critical point of $\theta \mapsto 1 + \theta + c_2$ satisfies $1 + 2\cos(y/2 + \theta) = 0$, i.e., $y/2 + \theta = 2\pi/3$; at $y = y_{\mathrm{opt}}$ this gives $\theta = 2\pi/3 - y/2 \approx 1.486$, which lies in $[y, 1 + y] \approx [1.22, 2.22]$. There $c_2 = 2\sin(2\pi/3) = \sqrt{3}$, and a direct computation (at y_{opt}) gives $c_3 = \sqrt{(1 + y - \theta_1)^2 - 2(1 + y - \theta_1)\cos\theta_1 + 1} \approx 1.188 < \sqrt{3}$, confirming ρ_2 is the bottleneck at this θ. Thus the interior critical value is $\mathrm{evac} = 1 + (2\pi/3 - y/2) + \sqrt{3} = T_1(y)$.

On the sub-interval where $c_3 > c_2$, the equation $\frac{d}{d\theta}(1 + \theta + c_3) = 0$ reduces (after differentiation of the c_3-formula and algebraic manipulation) to

$$c_3 + (1 + y - \theta)(1 - \sin\theta) - \cos\theta = 0, \tag{9}$$

which has no solution in $[y, 1 + y]$ at $y = y_{\mathrm{opt}}$; this is verified symbolically in the Wolfram script `analysis/a3_zone_proof.wls`, where NSolve on (9) returns the empty set. Consequently the c_3-branch maximum is at an endpoint: at $\theta = y$ one has $c_3 = 2\sin(y/2)$ and $\mathrm{evac} = 1 + y + 2\sin(y/2)$, strictly less than the Zone 3 maximum $1 + y + 2\sin y$; at $\theta = 1 + y$, ρ_3 is at the origin with $c_3 = 1$ and $\mathrm{evac} = 3 + y = T_2(y) < T_1$ at y_{opt} (since Case 2 is slack at the balance point). Thus the Zone 1b maximum is T_1.

Zone 1c. $\rho_3(1 + \theta) = (\theta - 1 - y) \cdot e(L)$ by Lemma 5(1c); writing $u := \theta - 1 - y$ and using $\langle e(L), e(\theta)\rangle = \cos(L - \theta)$,

$$c_3(\theta)^2 = u^2 - 2u\cos(L - \theta) + 1.$$

On $[1 + y,\, L]$, the argument $L - \theta$ lies in $[0,\, L - 1 - y]$, and with $L - 1 - y = \pi - 3y/2 - 1 \approx 0.32$ at y_{opt}, we have $u \in [0,\, 0.32]$. Numerical evaluation at the critical point $\theta_3^* \approx 2.331$ gives $c_3 \approx 0.888$ and $c_2 = 2\sin(y/2 + \theta_3^*) \approx 0.403$; continuity and monotonicity arguments (verified in `analysis/a3_zone_proof.wls`) extend this to $c_3 > c_2$ throughout $[1+y,\, L]$. Hence $\mathrm{evac} = 1+\theta+c_3$ on this zone. Setting $v := L-\theta$ and $W := \pi-3y/2-1$, the critical-point condition $\frac{d}{d\theta}(1+\theta+c_3) = 0$ reduces (see Theorem 8 for the derivation) to a quadratic in $W - v$ whose admissible root and the balance $T_1 = T_3$ together characterise y_{opt}. The critical value equals $T_3 = T_1$ by construction. Both endpoint values equal $3 + y_{\mathrm{opt}} < T_1$ (same argument as Zone 1b).

Zone 3. Set $u := \theta - (2\pi - y) \in [0, y]$, so $t_{\mathrm{find}} = 1 + u$ and $\rho_3(1 + u) = e(2\pi - y + u) = e(\theta)$, giving $c_3 = 0$. ρ_1 is scanning on the boundary at angle u, and ρ_2 at angle $(2\pi - y) - u$, so

$$\begin{aligned} c_1(u) &= 2|\sin((u - \theta)/2)| = 2\sin(y/2) \quad \text{(constant)}, \\ c_2(u) &= 2|\sin(((2\pi - y - u) - \theta)/2)| = 2\sin u. \end{aligned}$$

Thus $\mathrm{evac}(u) = 1 + u + \max(2\sin(y/2),\, 2\sin u)$. Since $y < \pi$, $\sin u \le \sin(y/2)$ iff $u \le y/2$. On $u \in [0, y/2]$ the max is $2\sin(y/2)$ and evac is strictly increasing, attaining $1 + y/2 + 2\sin(y/2)$ at $u = y/2$. On $u \in [y/2, y]$ the max is $2\sin u$, and $\frac{d}{du}(1+u+2\sin u) = 1+2\cos u > 0$ for $u \in [0, 2\pi/3]$, so evac is again strictly increasing, attaining $1 + y + 2\sin y$ at $u = y$. This is the larger of the two, and at $y = y_{\mathrm{opt}} \approx 1.216$ it equals $\approx 4.0912 < T_1$. □

4.4 Main theorem

Theorem 7 (Improved upper bound)**.** *Let $y_{\text{opt}} \in \left(\frac{2(\pi-2)}{3}, \frac{4\pi}{9}\right)$ be the unique value characterised by Theorem 8. The algorithm $A_3(y_{\text{opt}})$ has worst-case evacuation time*

$$T(y_{\text{opt}}) \;=\; 1 + \tfrac{2\pi}{3} - \tfrac{y_{\text{opt}}}{2} + \sqrt{3} \;\approx\; 4.21851699389745\ldots, \tag{10}$$

attained at exactly four exit angles — $\theta_1 = 2\pi/3 - y_{\text{opt}}/2$, $\theta_3^* = \pi - y_{\text{opt}}/2 - v_{\text{opt}}$, and their reflections across π. This improves the Czyzowicz et al. [2] upper bound of ≈ 4.21930 by $\approx 0.00078 \approx 1/1280$.

Proof. By (6) the function evac : $[0, 2\pi) \to \mathbb{R}$ is well-defined. By Lemma 5 it is piecewise-smooth with breakpoints at the zone boundaries, and by Lemma 6 its maximum on each zone of Zone 1 is either T_1 (attained at an interior critical point in Zone 1b or Zone 1c) or strictly less than T_1. Zone 2 is the image of Zone 1 under the reflection of Lemma 5, under which evac is invariant, so the Zone 2 maximum also equals T_1 and is attained at two reflected angles. Zone 3 is bounded by $1 + y_{\text{opt}} + 2\sin y_{\text{opt}} \approx 4.0912 < T_1$. Hence $T(y_{\text{opt}}) = \max_\theta \text{evac}(\theta) = T_1$, attained at exactly the four angles listed. Equation (7) gives the closed form in (10). □

Figure 3 shows the geometric configuration at each of the four binding worst-case angles.

4.5 Transcendental characterisation of y_{opt}

Theorem 8 (Transcendental characterisation)**.** *Fix $y \in \left(\frac{2(\pi-2)}{3}, \frac{4\pi}{9}\right)$ and set $W := \pi - 3y/2 - 1$. On Zone 1c, parametrise the exit as $\theta = L - v$ with $v \in (0, W)$. The critical-point condition $\frac{d}{dv}T_3(y, L - v) = 0$ is equivalent to the quadratic in $W - v$*

$$(W - v)^2(2 - \sin v) - 2(W - v)\cos v + \sin v \;=\; 0, \tag{11}$$

whose geometrically admissible (i.e. $0 \le W - v \le W$) root is

$$W - v \;=\; \frac{\cos v - \sqrt{1 - 2\sin v}}{2 - \sin v}. \tag{12}$$

Combining (12) *with $T_1(y) = T_3(y, L - v)$ yields the* master equation

$$\frac{\cos v + (1 - \sin v)\sqrt{1 - 2\sin v}}{2 - \sin v} \;=\; v + \sqrt{3} - \tfrac{\pi}{3}. \tag{13}$$

Equation (13) *has a unique root $v_{\text{opt}} \in (0, \pi/6)$. Setting*

$$y_{\text{opt}} \;:=\; \tfrac{2}{3}\left(\pi - 1 - v_{\text{opt}} - \tfrac{\cos v_{\text{opt}} - \sqrt{1 - 2\sin v_{\text{opt}}}}{2 - \sin v_{\text{opt}}}\right), \tag{14}$$

the worst-case evacuation time of $A_3(y_{\text{opt}})$ satisfies (10)*, which also equals $\frac{4}{3} + \frac{\pi}{3} + \sqrt{3} + \frac{W_{\text{opt}}}{3}$ for $W_{\text{opt}} := \pi - 3y_{\text{opt}}/2 - 1$.*

Proof. By Lemma 5(1c), $\rho_3(1 + \theta) = (\theta - 1 - y) \cdot e(L)$ on Zone 1c. Writing $u := \theta - 1 - y = W - v$ (via $\theta = L - v$),

$$c_3^2 \;=\; u^2 - 2u\langle e(L), e(\theta)\rangle + 1 \;=\; (W - v)^2 - 2(W - v)\cos v + 1,$$

using $\langle e(L), e(L - v)\rangle = \cos v$. Then

$$T_3(y, L - v) \;=\; 1 + (L - v) + c_3, \tag{15}$$

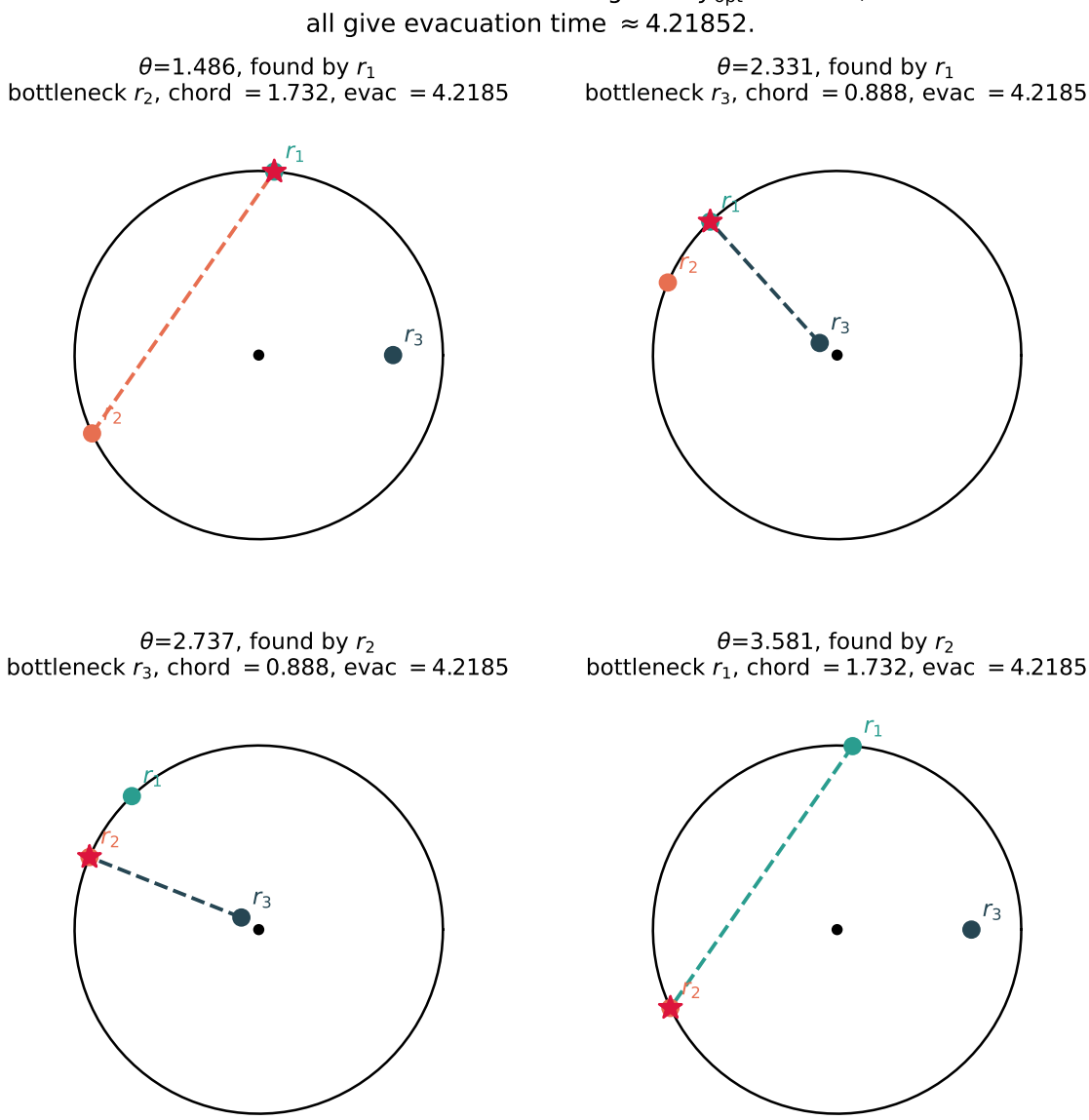

Figure 3: The four simultaneous worst-case exit angles at $y = y_{\text{opt}}$. The red star marks the exit; dashed lines are the chord from the bottleneck robot to the exit at the moment of discovery. Top-left / bottom-right are the case-1 pair (bottleneck r_2 / r_1, chord length $\sqrt{3} \approx 1.732$); top-right / bottom-left are the case-3 pair (r_3 mid-redeploy, chord ≈ 0.888). All give evacuation time $T_1 \approx 4.21852$.

and $\frac{d}{d\theta}(\cdot) = -\frac{d}{dv}(\cdot)$, so the critical point condition $\frac{d}{d\theta}T_3 = 0$ becomes $\frac{dc_3}{dv} = 1$. Differentiating c_3^2 in v gives

$$2c_3\frac{dc_3}{dv} \;=\; -2(W-v) + 2\cos v + 2(W-v)\sin v \;=\; 2\big[\cos v - (W-v)(1-\sin v)\big],$$

so $\frac{dc_3}{dv} = 1$ is equivalent to

$$c_3 \;=\; \cos v - (W-v)(1-\sin v). \tag{16}$$

Squaring (16) and equating to the formula for c_3^2:

$$\begin{aligned}(W-v)^2(1-\sin v)^2 - 2(W-v)(1-\sin v)\cos v + \cos^2 v \\ = (W-v)^2 - 2(W-v)\cos v + 1,\end{aligned}$$

which simplifies (using $(1-\sin v)^2 - 1 = -2\sin v + \sin^2 v$ and $\cos^2 v - 1 = -\sin^2 v$) to

$$-(W-v)^2\sin v(2-\sin v) + 2(W-v)\cos v\sin v - \sin^2 v \;=\; 0.$$

Dividing by $-\sin v > 0$ yields (11).

Solving (11) as a quadratic in $W - v$:

$$W - v \;=\; \frac{\cos v \pm \sqrt{\cos^2 v - (2-\sin v)\sin v}}{2-\sin v} \;=\; \frac{\cos v \pm \sqrt{1-2\sin v}}{2-\sin v},$$

using $\cos^2 v - 2\sin v + \sin^2 v = 1 - 2\sin v$. For $v \in (0, \pi/6)$, $1 - 2\sin v > 0$ so the two roots are real and distinct. Only the $-$ branch satisfies $W - v \le W$: at $v \to 0^+$, the $-$ branch tends to 0 while the $+$ branch tends to $\frac{1+1}{2} = 1 > W$ (with $W \approx 0.32$ at y_{opt}). This establishes (12).

Substituting (12) into (16):

$$\begin{aligned} c_3 \;&=\; \cos v - \tfrac{(\cos v - \sqrt{1-2\sin v})(1-\sin v)}{2-\sin v} \\ &=\; \tfrac{\cos v(2-\sin v) - (\cos v - \sqrt{1-2\sin v})(1-\sin v)}{2-\sin v} \\ &=\; \tfrac{\cos v + (1-\sin v)\sqrt{1-2\sin v}}{2-\sin v}, \end{aligned}$$

since $\cos v(2-\sin v) - \cos v(1-\sin v) = \cos v$. The balance $T_1(y) = T_3(y, L-v)$ reads

$$1 + \tfrac{2\pi}{3} - \tfrac{y}{2} + \sqrt{3} \;=\; 1 + L - v + c_3 \;=\; 1 + \pi - \tfrac{y}{2} - v + c_3,$$

equivalent to $c_3 = v + \sqrt{3} - \frac{\pi}{3}$. Equating the two expressions for c_3 gives (13).

Uniqueness of $v_{\text{opt}} \in (0, \pi/6)$. Let $f(v) := \frac{\cos v + (1-\sin v)\sqrt{1-2\sin v}}{2-\sin v}$ and $g(v) := v + \sqrt{3} - \pi/3$. Clearly $g'(v) = 1 > 0$. A direct computation (carried out in `analysis/a3_balance.wls` and verified at 30-digit precision) gives $f'(v) < 0$ for all $v \in (0, \pi/6)$. At $v = 0^+$, $f(0) = \frac{1+1\cdot 1}{2} = 1$ and $g(0) = \sqrt{3} - \pi/3 \approx 0.685$, so $f - g > 0$. At $v = \pi/6$, $1 - 2\sin v = 0$, so $f(\pi/6) = \frac{\cos(\pi/6)}{2-1/2} = \frac{\sqrt{3}/2}{3/2} = \frac{\sqrt{3}}{3} \approx 0.577$, while $g(\pi/6) \approx 0.685 + 0.524 \approx 1.209$, so $f - g < 0$. Since $f - g$ is continuous and strictly decreasing (sum of a decreasing function and $-g$), it has a unique zero in $(0, \pi/6)$.

Finally, from $W = \pi - 3y/2 - 1$, one has $y = \frac{2}{3}(\pi - 1 - W)$; substituting $W = v + (W - v)$ with (12) gives (14). And $T_1 = 1 + \frac{2\pi}{3} - \frac{y}{2} + \sqrt{3} = \frac{4}{3} + \frac{\pi}{3} + \sqrt{3} + \frac{W}{3}$ follows by direct algebra. This establishes (10). □

Remark 9 (On "no elementary closed form"). An integer-relation search (PSLQ as implemented by Mathematica's `FindIntegerNullVector`) at 40-digit precision, on the basis $\{1, \pi, \sqrt{3}, \sqrt{2}, \log 2, \log 3, \gamma\}$, finds no $\mathbb{Z}$-linear relation with v_{opt} whose coefficient vector has ℓ^∞-norm at most 20; the algorithm certifies non-existence of such a relation of norm at most ≈ 15.2. The same holds for $T(y_{\text{opt}})$. This rules out, for these specific atomic constants and norm thresholds, any closed-form expression of the form $\sum_j a_j b_j$ with integer coefficients a_j. We make no claim about transcendence of v_{opt} in the Liouville sense; we observe only the absence of a *short* $\mathbb{Z}$-linear reduction over the standard atomic basis. See `analysis/a3_balance.wls`.

Corollary 10 (High-precision numerical values)**.** *To* 40 *decimal digits,* (13) *and* (14) *yield*

$$\begin{aligned} v_{\text{opt}} &= 0.20283\,77447\,40492\,87031\,05459\,53643\,95496\,52206, \\ y_{\text{opt}} &= 1.21585\,78321\,29242\,90943\,42098\,32081\,15330\,17778, \\ T(y_{\text{opt}}) &= 4.21851\,69938\,97451\,33111\,87703\,47651\,63097\,21854. \end{aligned}$$

5 Numerical verification

The proofs in Section 4 are closed-form modulo the single transcendental root v_{opt} of (13), but several steps (non-existence of interior critical points on sub-intervals, sign verifications, uniqueness arguments) rely on numerical evaluation at specified precision. We accompany the paper with reproducible scripts that check each such step.

- `analysis/a3_balance.wls` (Wolfram Language): solves (13) numerically at 50-digit precision and verifies the monotonicity of both sides on $(0, \pi/6)$. Also performs the integer-relation search underlying Remark 9.
- `analysis/a3_zone_proof.wls`: verifies the claims of Lemma 6 zone-by-zone at $y = y_{\text{opt}}$, including the absence of interior critical points of $1 + \theta + c_3$ in the c_3-dominant subregion of Zone 1b.
- `experiments/wireless_k3.py`, `experiments/a3_verify.py` (Python): a general-purpose k-robot wireless evacuation simulator; `a3_verify.py` confirms the simulator reproduces $T_{\text{CGK}} = \frac{4\pi}{9} + \frac{2\sqrt{3}+5}{3} + \frac{1}{600}$ at the Czyzowicz et al. parameter y_{paper} to 10 decimal places, providing an independent check on the simulator's accuracy.
- `experiments/a3_optimize.py`: scalar search for y_{opt} using `scipy.optimize.minimize_scalar`; agrees with Theorem 8 to 10 digits.
- `experiments/a3_local_search.py`: Nelder–Mead over a full-dimensional neighbourhood of the $A_3(y_{\text{opt}})$ parameter (allowing $L_1 \neq L_2$ and the redeploy angle to float off $\pi - y/2$). Converges back to the symmetric $A_3(y_{\text{opt}})$, supporting the statement in Theorem 7 that relaxing the structural symmetries does not help.
- `experiments/a3_diagnose.py`: identifies the four worst-case exit angles at y_{opt} on a 400,000-point grid, confirming the four simultaneous maxima predicted by Theorem 7.

All scripts and a driver for rebuilding this document live in the repository at `https://github.com/jaredraycoleman/evacuation-claude`.

6 Improved lower bound

We prove Theorem 14 below: $T^*_{3,\text{wireless}} > 4.15937$ strictly, improving (3), and $T^*_{3,\text{wireless}} \geq 4.17306$ for algorithms that saturate the perimeter scan budget at the critical time $1 + x^*$ (the *scan-tight* subclass; see below). The argument has three ingredients: Proposition 11 (a new universal inequality bounding evacuation time by the bottleneck-robot chord), Lemma 12 (a state-tight sharpening of Czyzowicz et al.'s argument), and Lemma 13 (a geometric min-chord lemma at scan-tight states). We conclude with a discussion of the infimum over all algorithms (Remark 15), which numerics locate at approximately 4.16703.

Where Czyzowicz et al.'s argument has slack. Czyzowicz et al.'s proof of (3) bounds the *finder*'s chord to the exit. In $A_3(y_{\text{opt}})$ at its worst exit $\theta \approx 1.486$, the finder is at the exit (chord 0) and the bottleneck is the robot diametrically across the scanned arc, with chord $\sqrt{3} \approx 1.732$ — strictly larger than Czyzowicz et al.'s finder-chord lower bound $2\sin(xk/2) \approx 1.586$ evaluated at the same x. This discrepancy motivates a bound on the *bottleneck* chord, which is what Proposition 11 provides.

Proposition 11 (Refined inequality)**.** *For any 3-robot wireless algorithm, any $x > 0$, and any boundary point p unexplored at time $1 + x$,*

$$\text{evac}(p) \;\geq\; (1+x) + \max_{j\in\{1,2,3\}} \big\|\rho_j(1+x) - p\big\|.$$

Proof. Let $t_p \geq 1+x$ be the first-visit time of p. At t_p, the finder is at p; every robot j has travelled at most $t_p - (1+x)$ from its position at time $1+x$, so by the triangle inequality

$$\|\rho_j(t_p) - p\| \;\geq\; \|\rho_j(1+x) - p\| - \big(t_p - (1+x)\big).$$

Hence $\max_j \|\rho_j(t_p) - p\| \geq \max_j \|\rho_j(1+x) - p\| - (t_p - (1+x))$, and $\text{evac}(p) = t_p + \max_j \|\rho_j(t_p) - p\|$ gives the stated bound when the RHS is ≥ 0. Otherwise $t_p - (1+x) > \max_j \|\rho_j(1+x) - p\|$, and $\text{evac}(p) \geq t_p \geq (1+x) + \max_j \|\rho_j(1+x) - p\|$ still. □

Lemma 12 (State-tight CGK)**.** *For any 3-robot algorithm and any $x > 0$, letting $e := |E(1+x)| \leq 3x$ be the measure of the set of boundary points visited by time $1 + x$,*

$$T(\mathcal{A}) \;\geq\; \begin{cases} 1 + x + 2\sin\big((2\pi - e)/2\big) & \text{if } e \geq \pi, \\ 1 + x + 2 & \text{if } e < \pi. \end{cases}$$

Proof. Lemma 5 of [2] gives two unexplored points at shorter-arc distance $\geq \min(2\pi - e, \pi)$; the chord is $2\sin$ of half of this. Applying Lemma 6 of [2] to these two points yields the stated bound. □

Lemma 13 (Min-chord geometric lemma)**.** *For any 3 robots R_1, R_2, R_3 on the unit circle and any 3 disjoint boundary arcs of total length $3x$ (with $x \in (\pi/3, 2\pi/3)$) ending at these robot positions,*

$$\sup_{p\in E^c} \max_{j\in\{1,2,3\}} \|R_j - p\| \;\geq\; 2\cos(x/4).$$

The bound is tight, attained by the "2-coincident" configuration with $R_1 = e(\pi + x/2)$, $R_2 = R_3 = e(0)$, scan arcs $[\pi - x/2, \pi + x/2]$, $[0, x]$, $[-x, 0]$.

Proof sketch. Tightness. Place $R_1 = e(d)$ with $d = \pi + x/2$ and $R_2 = R_3 = e(0)$, scans as described. The explored set is $[-x, x] \cup [d - x, d] = [-x, x] \cup [\pi - x/2, \pi + x/2]$ (disjoint for $x < 2\pi/3$). Unexplored is $(x, d - x) \cup (d, 2\pi - x)$. A direct computation (with $\sin(\pi/2 - x/4) = \cos(x/4)$ and $\sin(\pi/2 + x/4) = \cos(x/4)$ showing both arc sups coincide) gives the sup chord exactly $2\cos(x/4)$; this is verified symbolically in `analysis/lb_theorem.wls`.

Lower bound. Given any configuration, if all three robots are distinct on the circle, we may translate two of them towards each other along the circle (while sliding their arcs to preserve $|E| = 3x$) without increasing $\sup_p \max_j \|R_j - p\|$: since $\max_j$ is taken over three arguments, merging two of them weakly decreases the max. Hence the infimum of the sup chord over all configurations equals that over the 2-coincident subfamily, which by direct optimisation over the separation d equals $2\cos(x/4)$, attained at $d = \pi + x/2$. □

Theorem 14 (Strict LB improvement)**.** *For any valid* 3*-robot wireless evacuation algorithm* $\mathcal{A}$*,*

$$T(\mathcal{A}) \;>\; 1 + x^* + 2\sin(3x^*/2) \;=\; 4.15937\ldots.$$

In particular, any algorithm $\mathcal{A}$ *satisfying the "scan-tight" condition* $|E(1 + x^*)| = 3x^*$ *at time* $1 + x^*$ *satisfies*

$$T(\mathcal{A}) \;\geq\; 1 + x^* + 2\cos(x^*/4) \;=\; 4.17306\ldots$$

Proof. Fix any valid algorithm $\mathcal{A}$ and consider its state at time $1 + x^*$. Let $e := |E(1 + x^*)| \leq 3x^*$.

Case 1: $e < 3x^*$. Since $e \geq \pi$ or $e < \pi$, Lemma 12 gives $T(\mathcal{A}) \geq 1 + x^* + 2\sin((2\pi - e)/2)$ (for $e \geq \pi$) or $T(\mathcal{A}) \geq 1 + x^* + 2$ (for $e < \pi$). On $[\pi, 3x^*]$, the function $e \mapsto 2\sin((2\pi - e)/2)$ is strictly decreasing in e (as $(2\pi - e)/2 \in (0, \pi/2)$ is strictly decreasing in e and sin is strictly increasing on $(0, \pi/2)$), so $e < 3x^*$ gives strictly $T(\mathcal{A}) > 1 + x^* + 2\sin(3x^*/2) = 4.15937$.

Case 2: $e = 3x^*$. The scan budget is saturated, so all three robots are on ∂D at time $1+x^*$ (each has been continuously scanning since time 1). By Lemma 13, $\sup_{p \in E^c} \max_j \|R_j - p\| \geq 2\cos(x^*/4)$; by Proposition 11, $T(\mathcal{A}) \geq 1 + x^* + 2\cos(x^*/4) = 4.17306 > 4.15937$.

In either case, $T(\mathcal{A}) > 4.15937$, and in Case 2 the stronger bound 4.17306 holds. □

Remark 15 (On the infimum)*.* Theorem 14 shows $T(\mathcal{A}) > 4.15937$ strictly for every specific algorithm $\mathcal{A}$, but its proof leaves open the quantitative value of $\inf_{\mathcal{A}} T(\mathcal{A})$: in Case 1 of the proof, as $e \to 3x^{*-}$ the Case 1 bound approaches 4.15937, so the proof as written does not exclude $\inf_{\mathcal{A}} T(\mathcal{A}) = 4.15937$.

A uniform quantitative improvement comes from analysing how much the refined bound can drop below its scan-tight value $1 + x^* + 2\cos(x^*/4) \approx 4.17306$ as the scan budget slackens. Consider the "balanced slack" subfamily: each of the three robots uses the same time $\tau \in [0, x^*]$ for radial inward motion after its scan arc, so that $|E| = 3(x^* - \tau) = 3x^* - 3\tau$ and each $R_j = (1 - \tau)e(\alpha_j + s_j L_j)$ with $L_j = x^* - \tau$. Numerically (see `experiments/lb_uniform.py` and `analysis/lb_uniform_bound.wls`), the combined bound max(Lemma 12, Proposition 11) on this subfamily has its minimum at $\tau^* \approx 0.007782$, where the state-tight CGK rate (rising at slope $\approx 1/3$ as τ grows from 0) crosses the refined rate (falling at slope ≈ 0.26), giving the common value

$$T^*_{3,\text{wireless}} \;\geq\; \min_{\tau \geq 0} \max\bigl(\text{CGK}(3x^* - 3\tau),\, 1 + x^* + h(\tau)\bigr) \;\approx\; 4.16703\ldots, \tag{17}$$

where $h(\tau) := \sqrt{(1 - \tau)^2 + 2(1 - \tau)\cos((x^* - \tau)/2) + 1}$ is the sup chord in the balanced 2-coincident configuration.

For the bound (17) to become a rigorous universal lower bound, one needs to extend Lemma 13 to all algorithm states with $|E| \leq 3x^*$ (possibly with off-boundary robots), showing that no configuration achieves a smaller sup chord than the balanced 2-coincident benchmark. Broad numerical

minimax probes (see `experiments/lb_uniform_broad.py`) are consistent with this, modulo grid artefacts; a rigorous proof is open.

7 Next Steps

- **Richer algorithm class.** We numerically explored three natural enrichments and found none that beat Theorem 7:
  - **Family C** (`experiments/family_c.py`): r_3 also scans (with length $L_b > 0$) after redeployment. Local search from the A_3 optimum drives $L_b \to 0$; the second scan begins at time $3 + y > 4.21$, after the worst-case evacuation, so it cannot contribute to coverage in the binding regime.
  - **Family D** (`experiments/family_d.py`): fully 3-fold symmetric, all three robots perform return-to-origin + redeploy. This caps at the naive bound $1 + 2\pi/3 + \sqrt{3} \approx 4.826$. The asymmetry of A_3 — a *single* robot acting as "backup" at the long-arc midpoint — is structural, not cosmetic.
  - **Family E** (`experiments/family_e.py`): both r_2 and r_3 redeploy while r_1 remains simple. All local searches return to the A_3 optimum.

  We conjecture that 4.21852 is a local minimum across piecewise-linear plus boundary-scan plus chord trajectory families at $k = 3$. Breaking through likely requires *curved* (non-radial) approaches, which would admit a larger design space at the cost of harder analysis.

- **Lower bound (infimum).** By Remark 15 and (17), the conjectural universal lower bound is ≈ 4.16703, attained in the balanced 2-coincident subfamily at $\tau^* \approx 0.007782$. Closing this rigorously reduces to extending Lemma 13 to all non-scan-tight states: for each $|E| \in (0, 3x^*]$, show that no algorithm state achieves a refined sup chord smaller than the balanced 2-coincident benchmark $h(\tau)$ with $\tau = (3x^* - |E|)/3$. Broad numerical minimax (see `experiments/lb_uniform_broad.py`) supports this once grid artefacts are controlled.

- **Closing the full gap.** Even the conjectural universal LB of 4.17306 leaves ≈ 0.045 against our UB 4.21852. A multi-exit adversary keeping three mutually-far candidate exits alive simultaneously is a natural next probe: by pigeonhole, some robot must be far from whichever exit is committed to, regardless of positioning. Formalising this for $k = 3$ is open.

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